## Absolute error

**1**

**a** 0.5 kg **b** 0.5 g **c** $5

**2** $\frac{1}{2}$ mL

**3**

**a** 0.05 mL **b** 0.5 mm **c** 0.05 g **d** 0.5 s

**e** 0.005 kg **f** 0.05 m **g** 0.005 °C

**4** We do not know what place value the number has been rounded to.

## The limits of accuracy

**5**

**a** 1 cm **b** 90 cm

**c** 0.5 cm **d** **i** No **ii** No **iii** No **iv** Yes

**6**

**a** 10 kg **b** 5 kg

**c** Lower bound: 65 kg, upper bound: 75 kg

**7**

**a** 50 km/h **b** 5 km/h **c** 55 km/h

**8**

**a** **i** 13.455 km **ii** 13.445 km **b** **i** 6.45 km **ii** 6.35 km

**c** **i** 5.5 m **ii** 4.5 m **d** **i** 14.005 kg **ii** 13.995 kg

**e** **i** 4884.5 m **ii** 4883.5 m **f** **i** $52.50 **ii** $47.50

**9** 19.995 m

**10** 2.05 m

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**11**

**a** Maximum: 175, minimum: 165

**b** Maximum: 3050, minimum: 2950

## Further calculations (Extended)

**12**

**a** 26.25

**b** 0.6

**13**

**a** 13.85 m/s

**b** 13.95 m/s

**14**

**a** 153.9 $\text{cm}^2$

**b** 132.7 $\text{cm}^2$, 176.7 $\text{cm}^2$

**c** 22.8 $\text{cm}^2$

**d** 14.8%

**15**

**a** Upper bound: 272.6925 $\text{m}^2$, lower bound: 269.3925 $\text{m}^2$

**b** Upper bound: 66.2 m, lower bound: 65.8 m

**16**

**a** Lower bound: 9.5 cm, upper bound: 10.5 cm

**b** 100 $\text{cm}^2$

**c** Lower bound: 90.25 $\text{cm}^2$, upper bound: 110.25 $\text{cm}^2$

**d** 1000 $\text{cm}^3$

**e** Lower bound: 857.375 $\text{cm}^3$, upper bound: 1157.625 $\text{cm}^3$

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